AAE1001 Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering

Week 7 (Introduction to Path Planning)

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Introduction to A* Path Planning Algorithm

Daily Path Planning

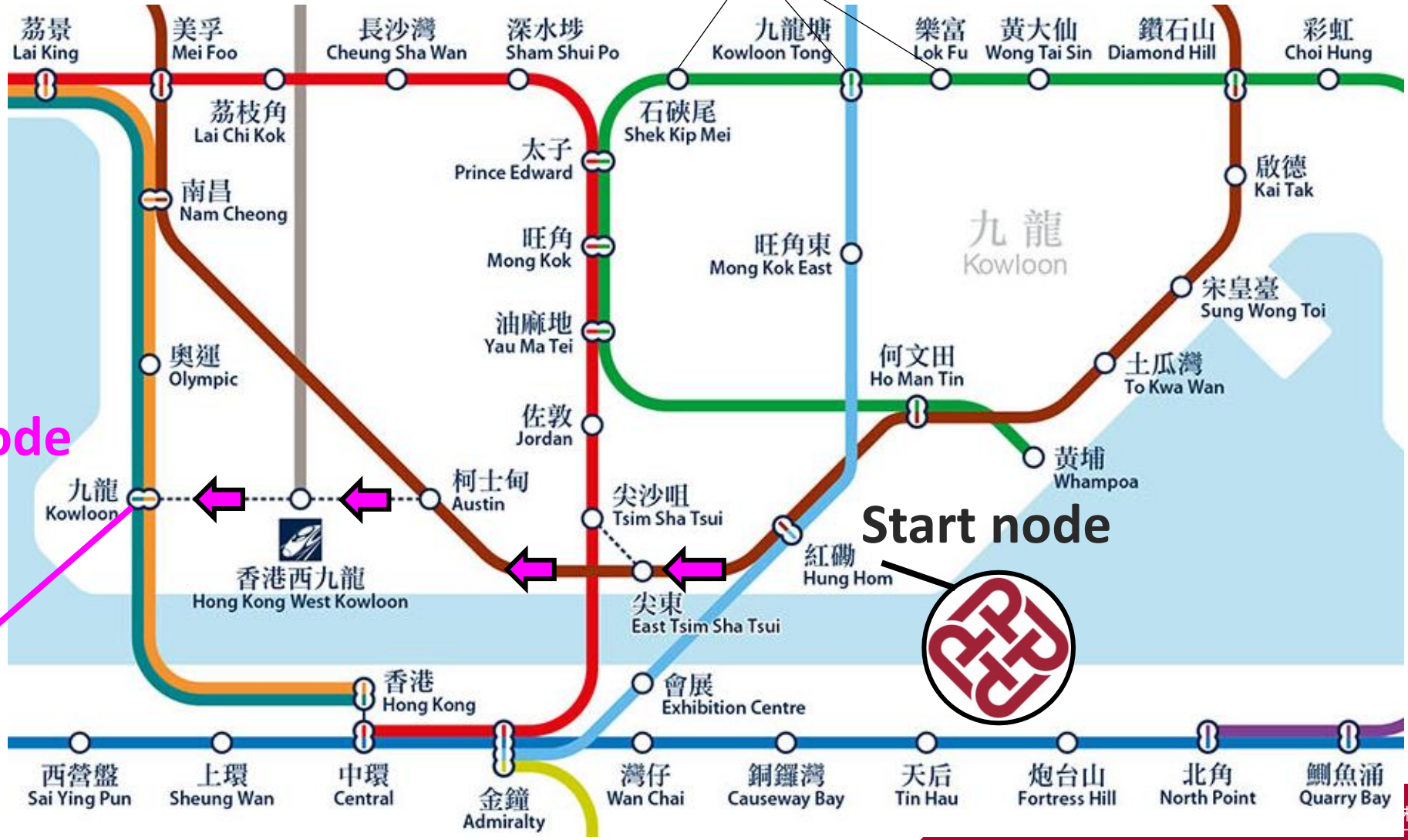
MTR stations – available **node** to go



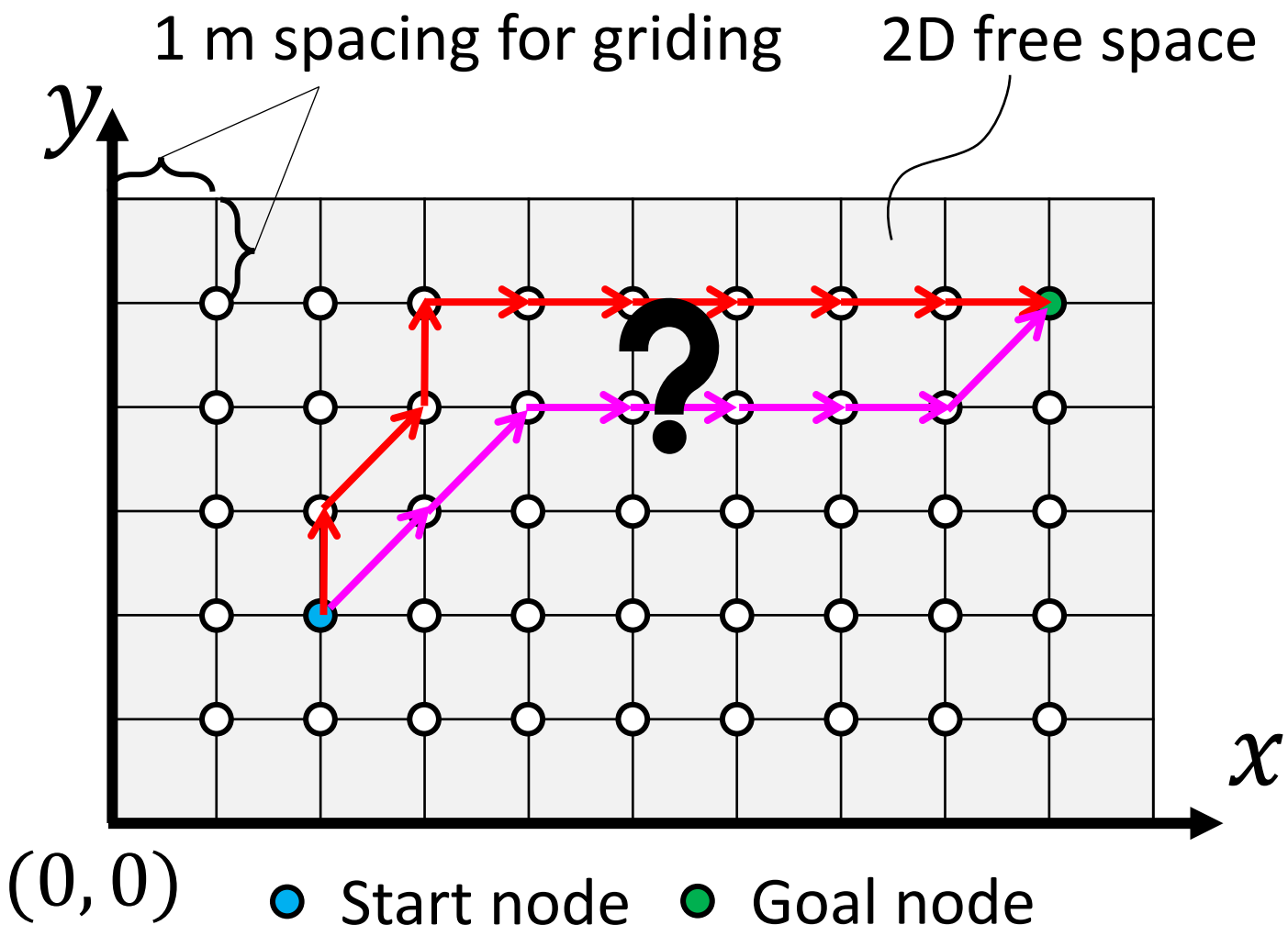


Goal node





Definition of Path Planning



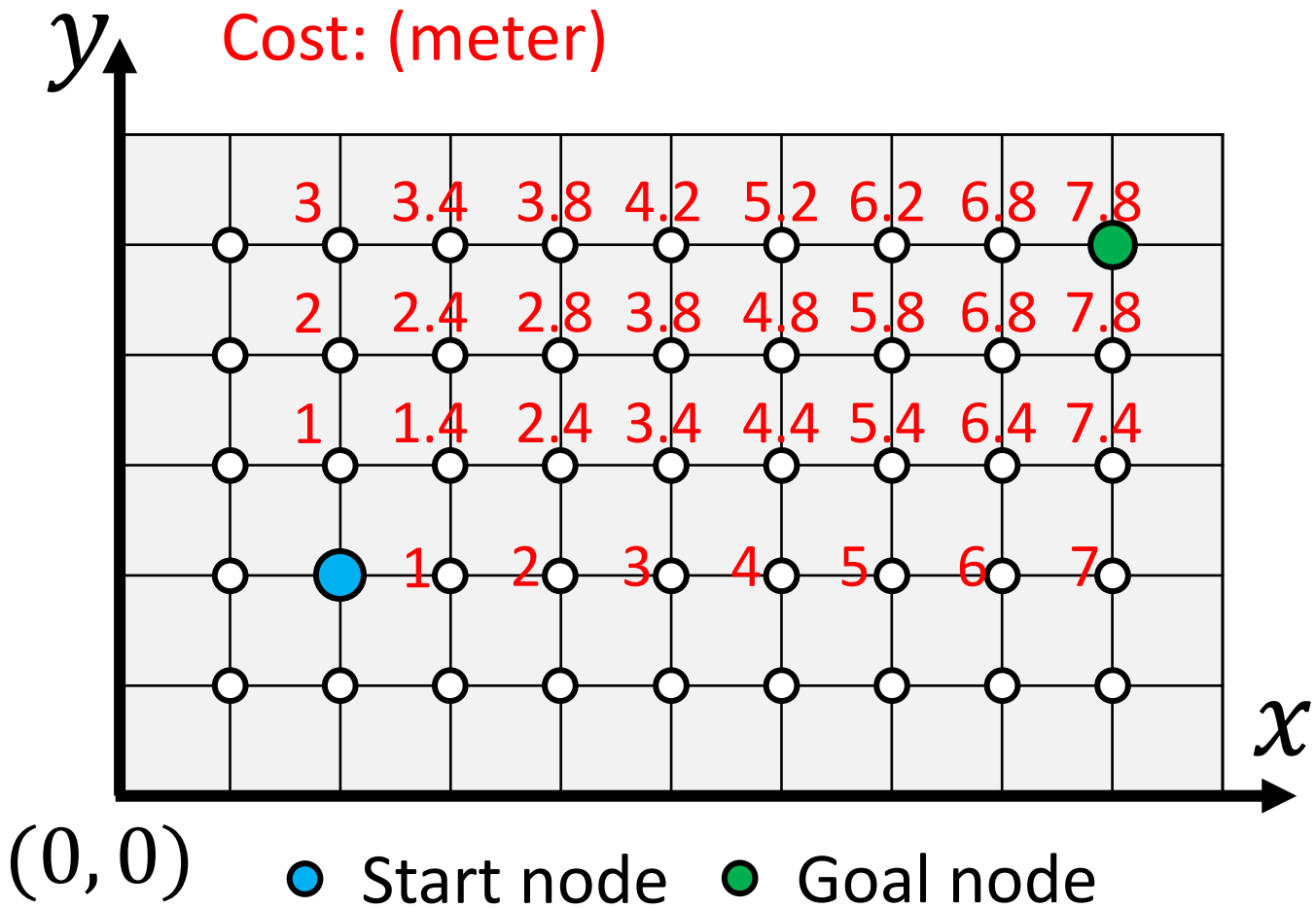
- **Node** — All potential position you can go across with a unique position (x, y)

- **Search space** — A collection of nodes, like all board positions of a board game.

- **Objective of path planning**— Find the shortest routes with smallest cost from start node to goal node.

Which one is better?

Path Planning by Checking All Available Nodes



Test each possible nodes one-by-one from Start node

⇩

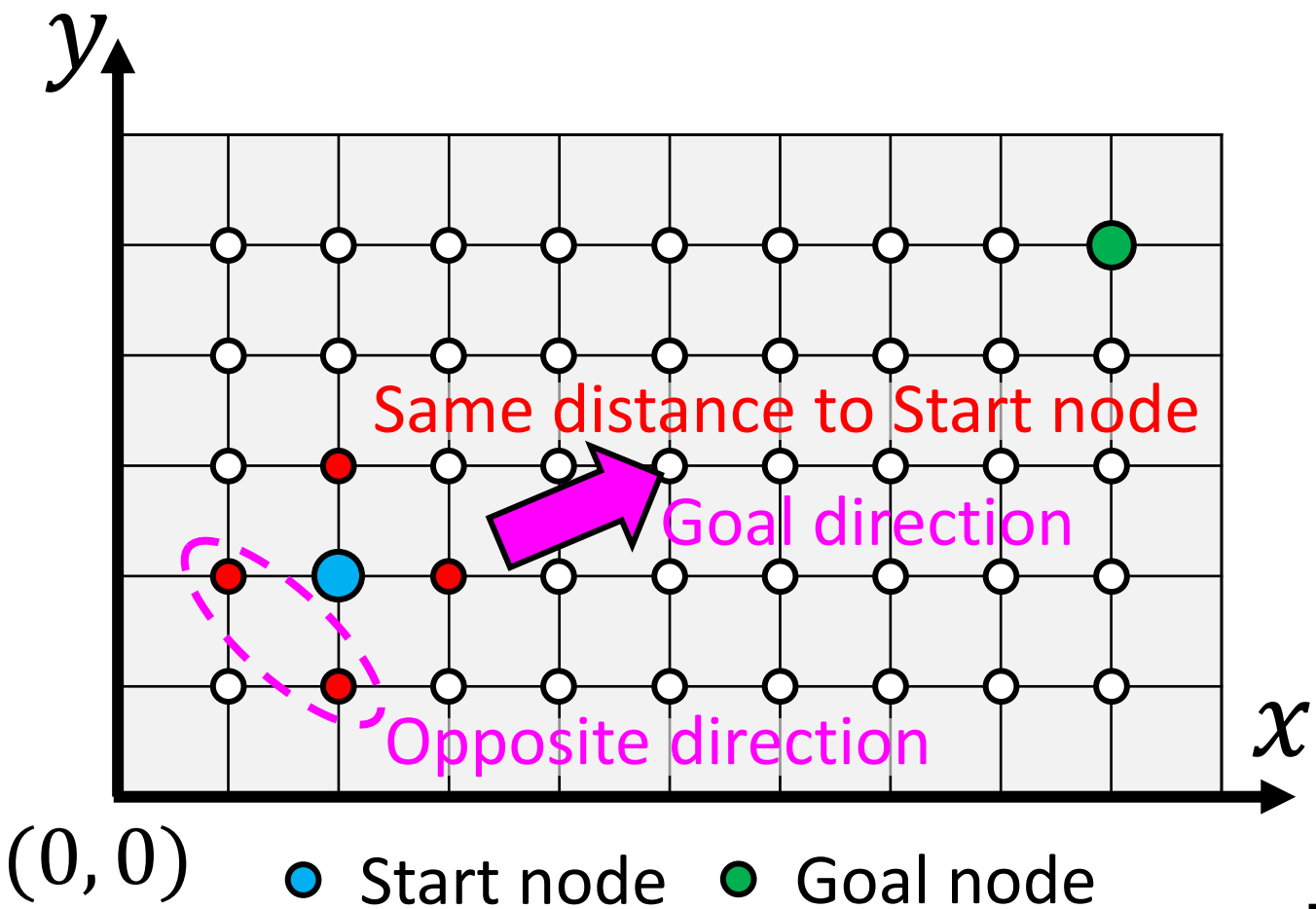
Record its shortest path between Start node

Reach Goal ⇩

Retrieve the shortest path (Dijkstra's algorithm)

Need higher efficiency!

Path Planning by Checking **NOT** All Available Nodes



Test each possible nodes one-by-one from Start node

⇩

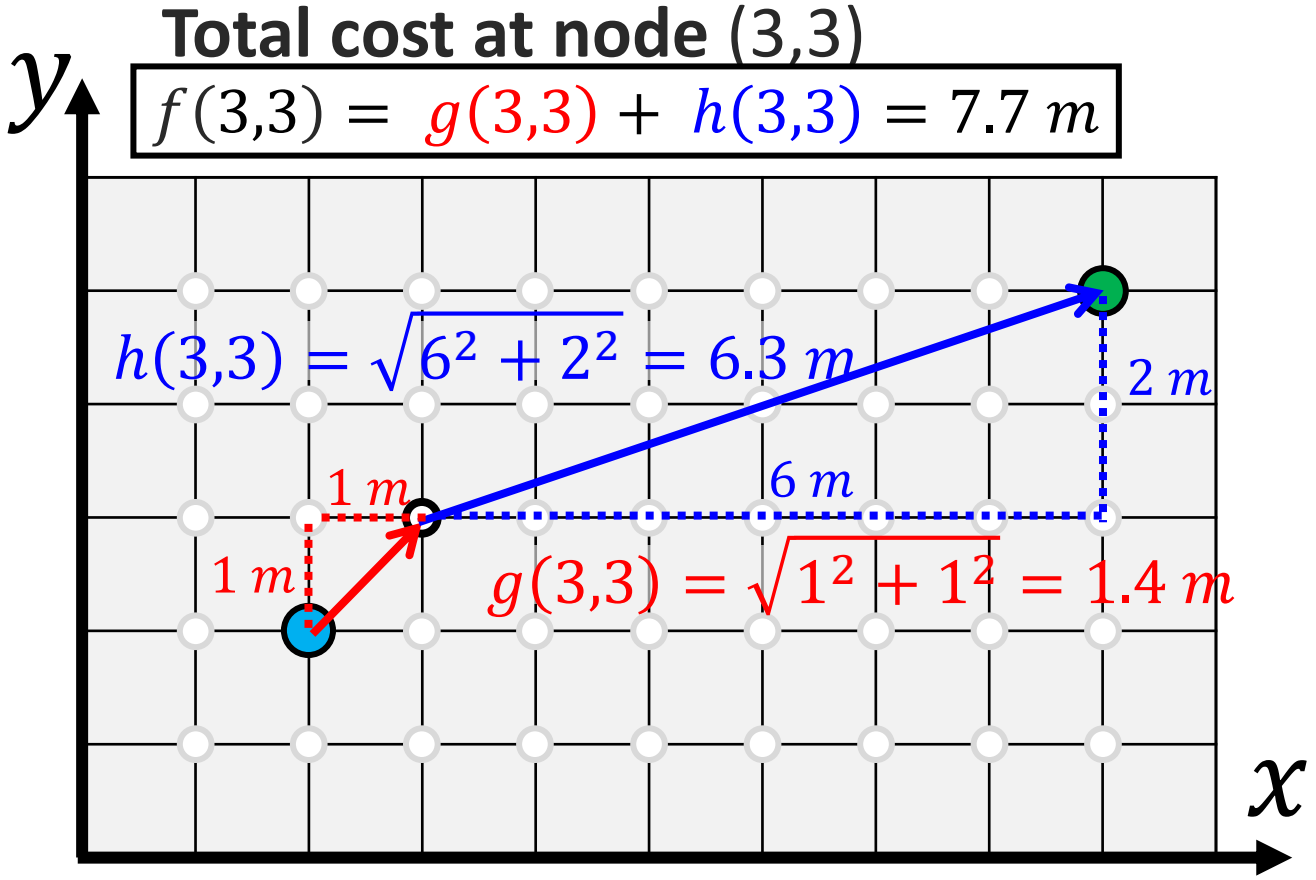
Record its shortest path between Start node



We also know a rough direction to Goal!

⇨ **A-star Search Algorithm**

Path Planning by A-star (A*) Search Algorithm



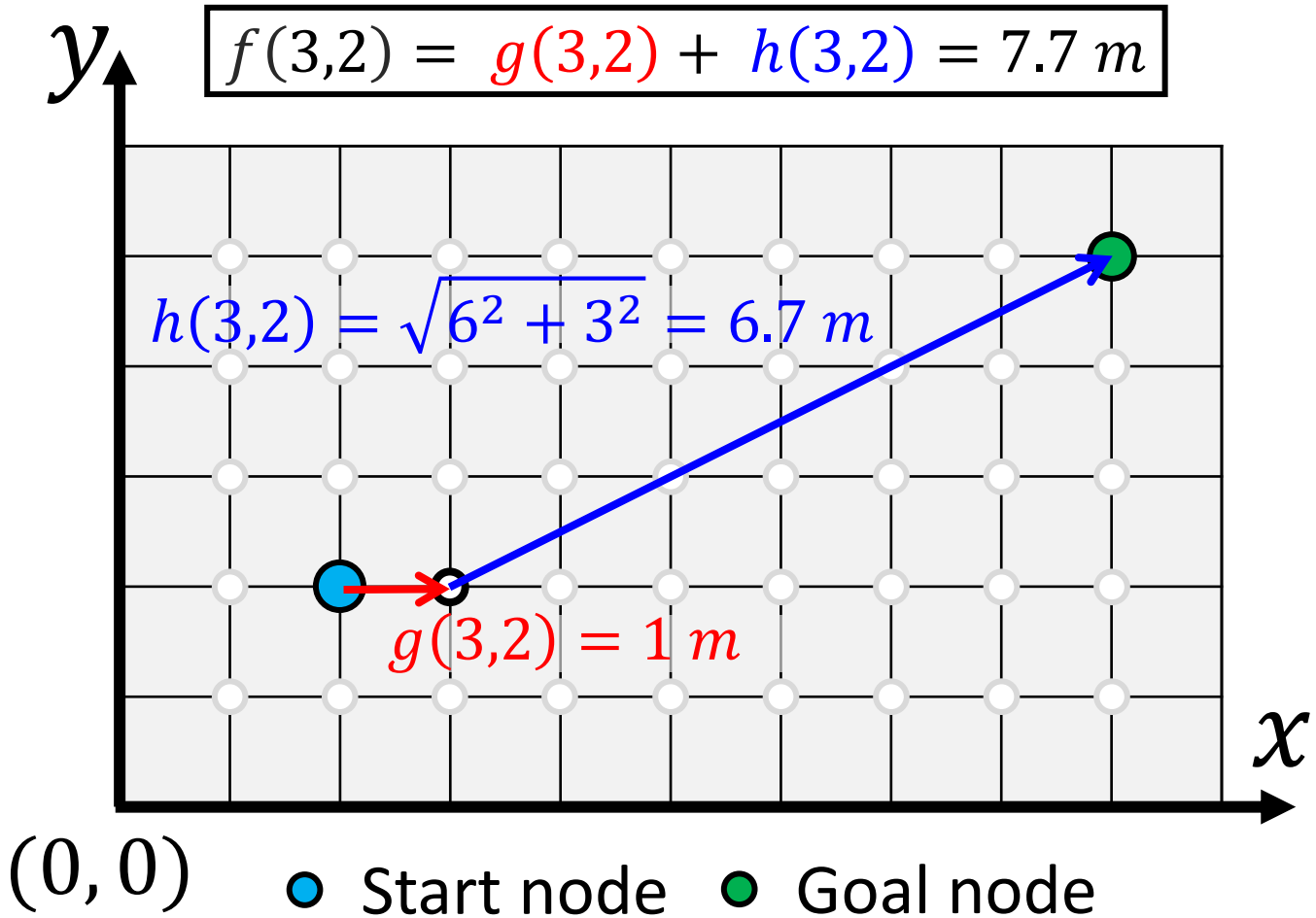
Definition of cost:

$g(x,y)$ — this represents the ***exact cost*** of the path **from** the **Start** node to node (x,y)

$h(x,y)$ — this represents the **heuristic** ***estimated cost*** from node (x,y) **to** the **Goal** node

$f(x,y) = g(x,y) + h(x,y)$ — **total** cost of a neighboring node (x,y)

A-star (A*) Path Planning – Cost at Node (3,2)



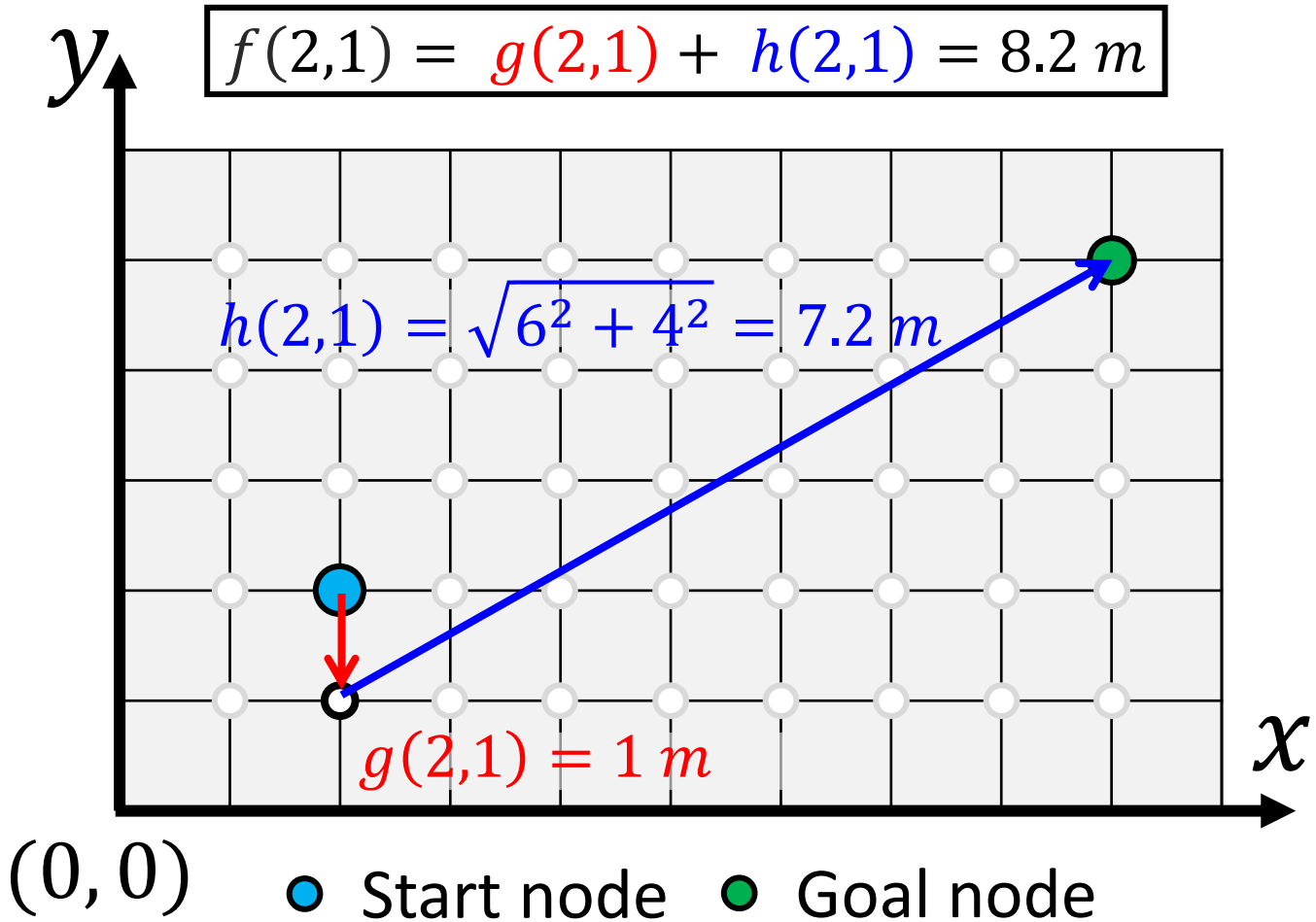
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$h(x,y)$ — this represents the heuristic ***estimated cost*** from node (x,y) **to** the **Goal** node

$f(x,y) = g(x,y) + h(x,y)$ — **total** cost of a neighboring node (x,y)

A-star (A*) Path Planning – Cost at Node (2,1)



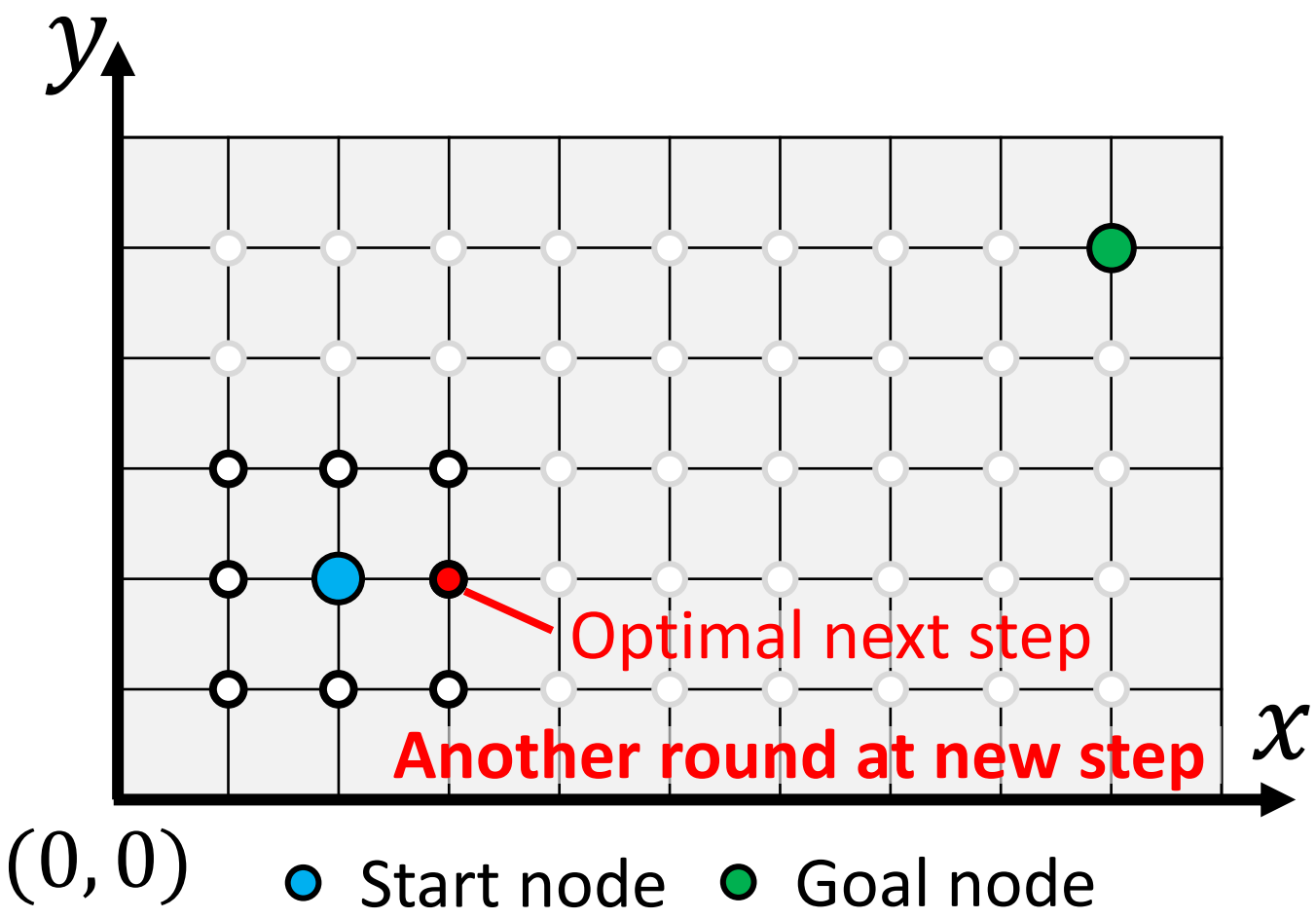
Definition of cost:

$g(x,y)$ — this represents the ***exact cost*** of the path **from** the **Start** node to node (x,y)

$h(x,y)$ — this represents the heuristic ***estimated cost*** from node (x,y) **to** the **Goal** node

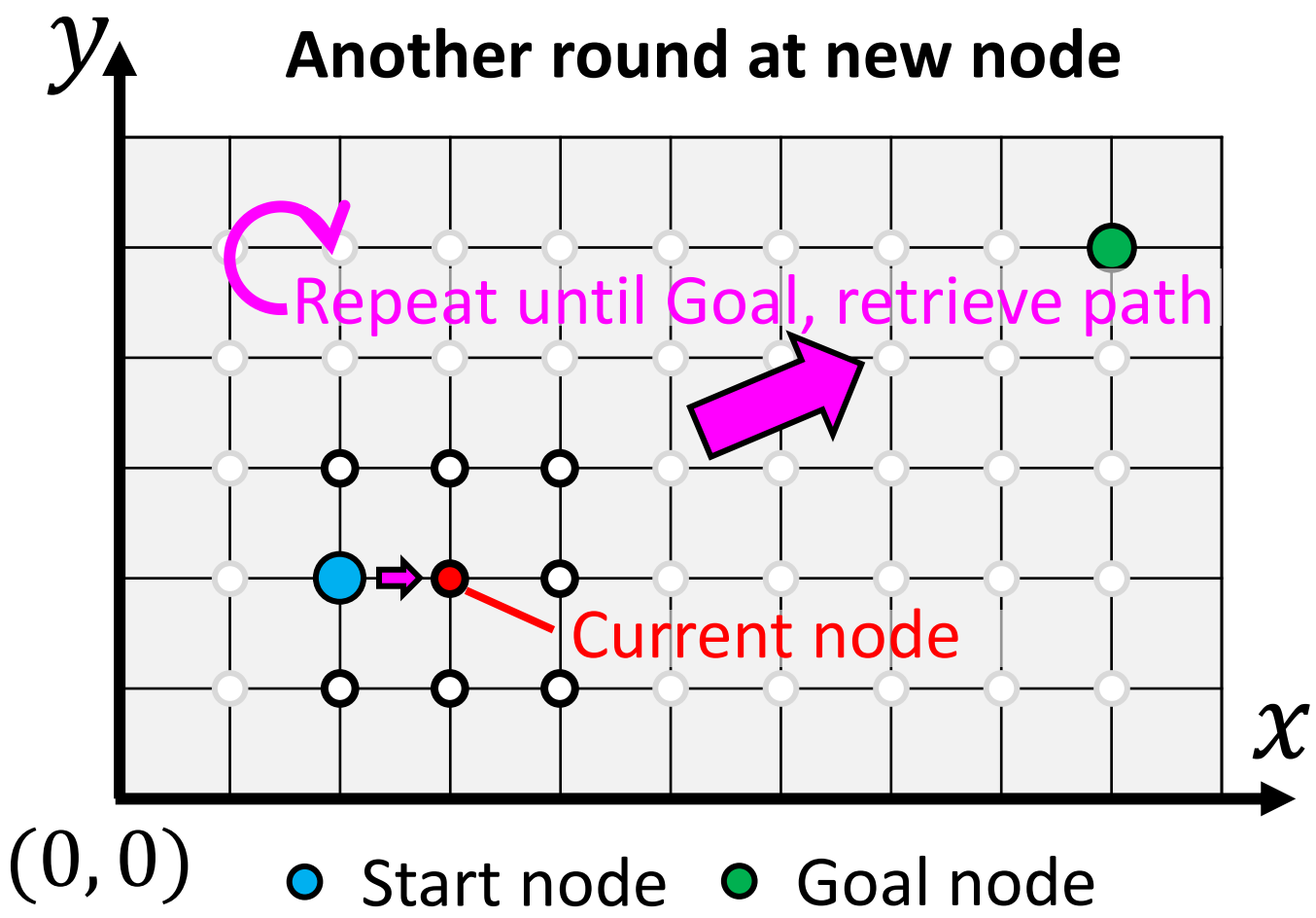
$f(x,y) = g(x,y) + h(x,y)$ — **total** cost of a neighboring node (x,y)

Total Costs at Neighbouring Nodes



Node ID	*Exact cost from Start* $g(x,y)$	*Estimated cost to Goal* $h(x,y)$	*Total cost* $f(x,y)$
(1,1)	1.4	8.9	10.3
(1,2)	1	8.5	9.5
(1,3)	1.4	8.2	9.6
(2,1)	1	8.1	9.1
(2,3)	1	7.3	8.3
(3,1)	1.4	7.2	8.6
(3,2)	1	6.7	7.7
(3,3)	1.4	6.3	7.7

Total Costs at Neighbouring Nodes – Round 2

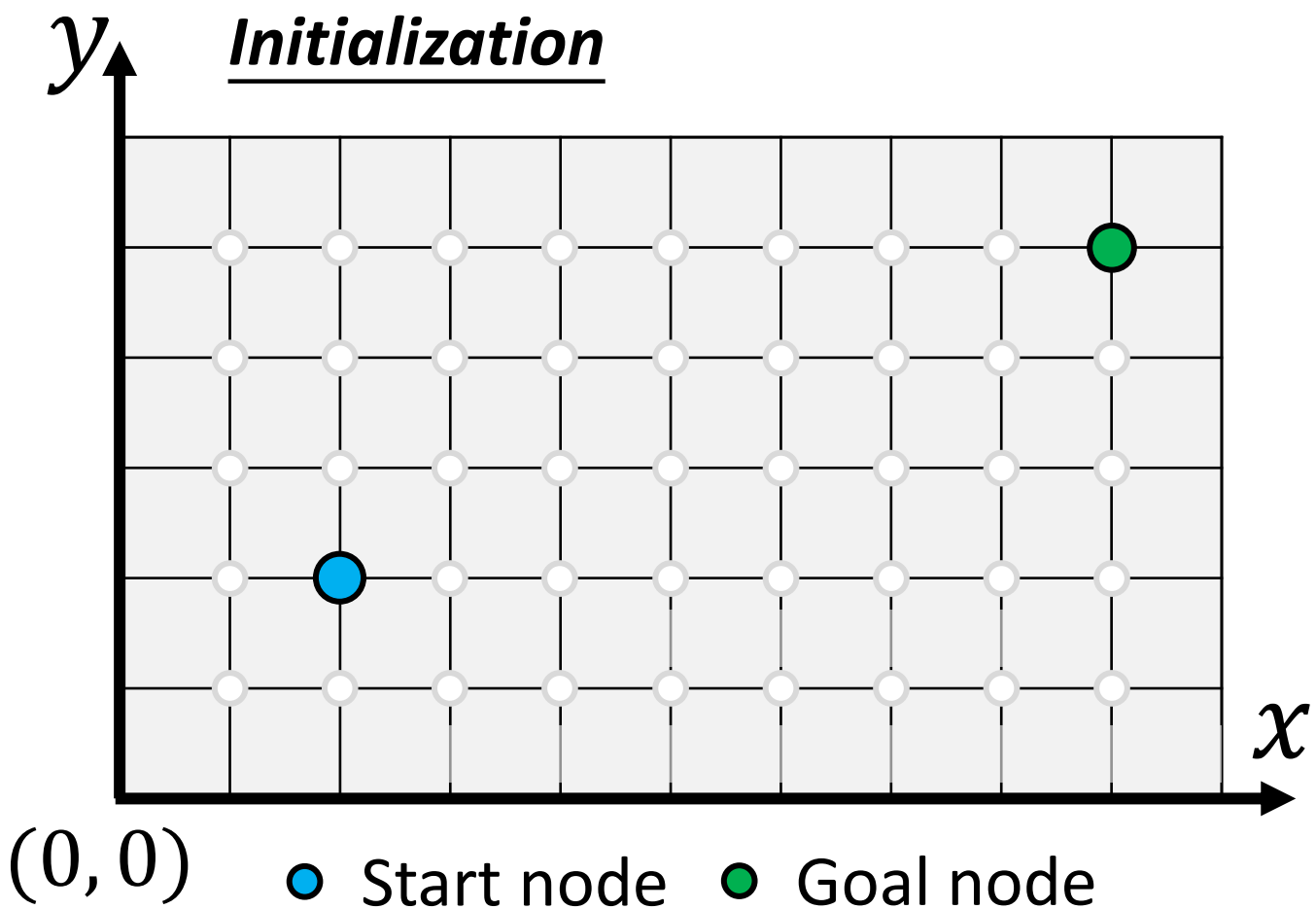


Exact cost from Start *Estimated cost to Goal* *Total cost*

Node ID	$g(x,y)$	$h(x,y)$	$f(x,y)$
(2,1)			
(2,2)			
(2,3)			
(3,1)			
(3,3)			
(4,1)			
(4,2)			
(4,3)			

How does computer search step by step and got path?

A-star (A*) Path Planning – Record Path



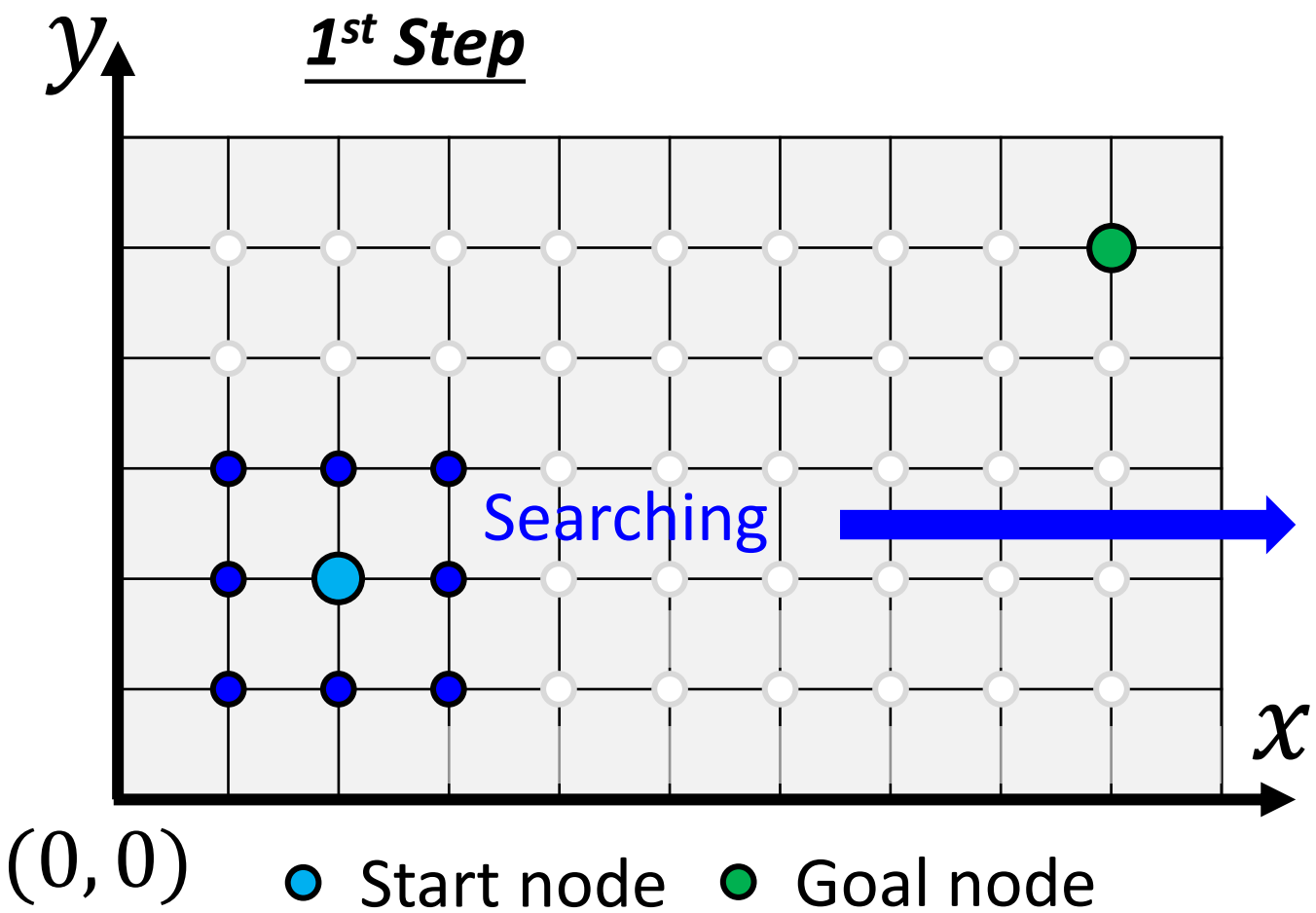
Open List
(searched nodes)

Node	f	Source

Close List
(arrived nodes)

Node	f	Source
Start	-	-

A-star (A*) Path Planning – Record Path



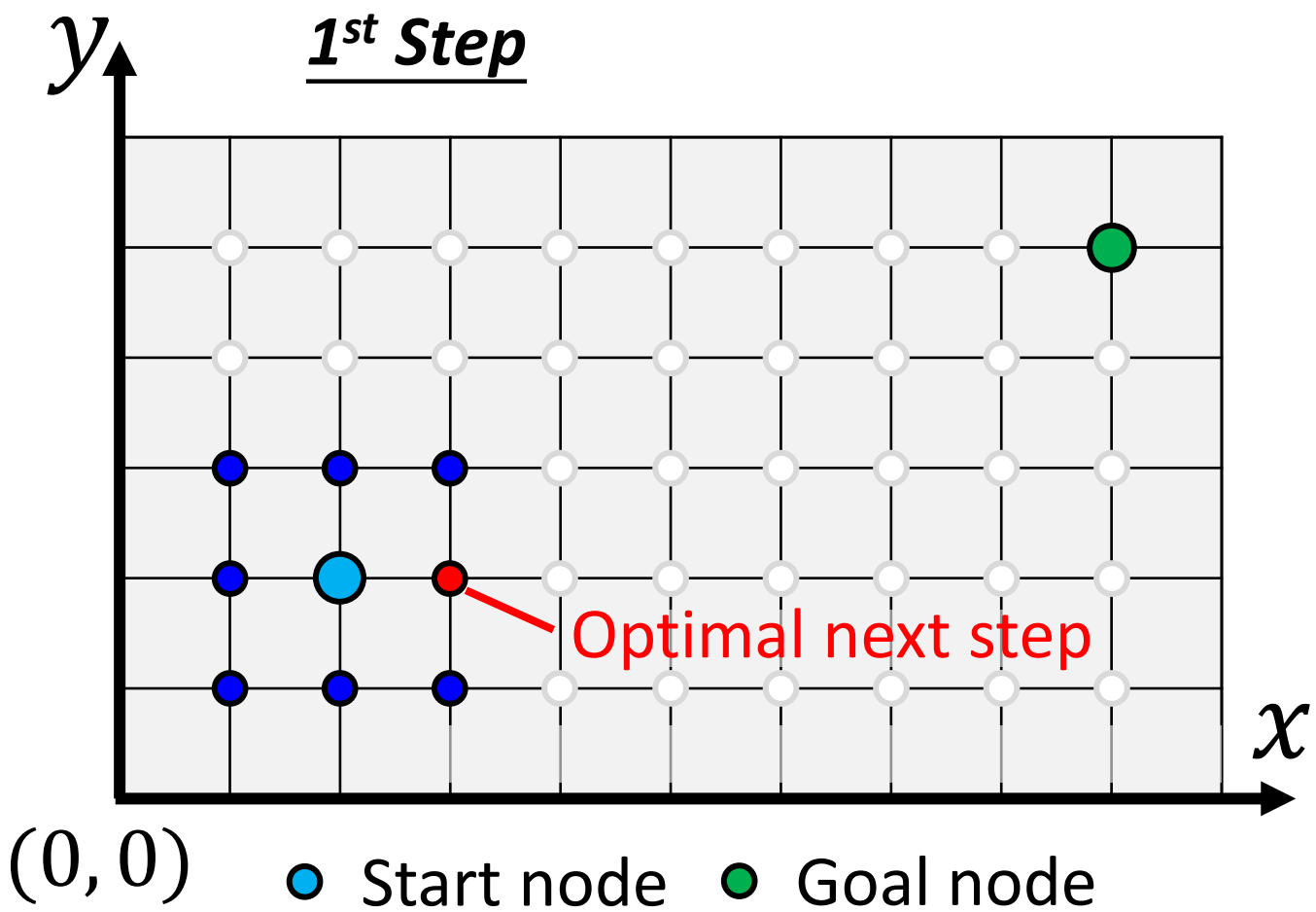
Open List *(searched nodes)*

Node	f	Source
(1,1)	8.9	(2,2)
(1,2)	8.5	(2,2)
(1,3)	8.2	(2,2)
(2,1)	8.1	(2,2)
(2,3)	7.3	(2,2)
(3,1)	7.2	(2,2)
(3,2)	6.7	(2,2)
(3,3)	6.3	(2,2)

Close List *(arrived nodes)*

Node	f	Source
Start	-	-

A-star (A*) Path Planning – Record Path



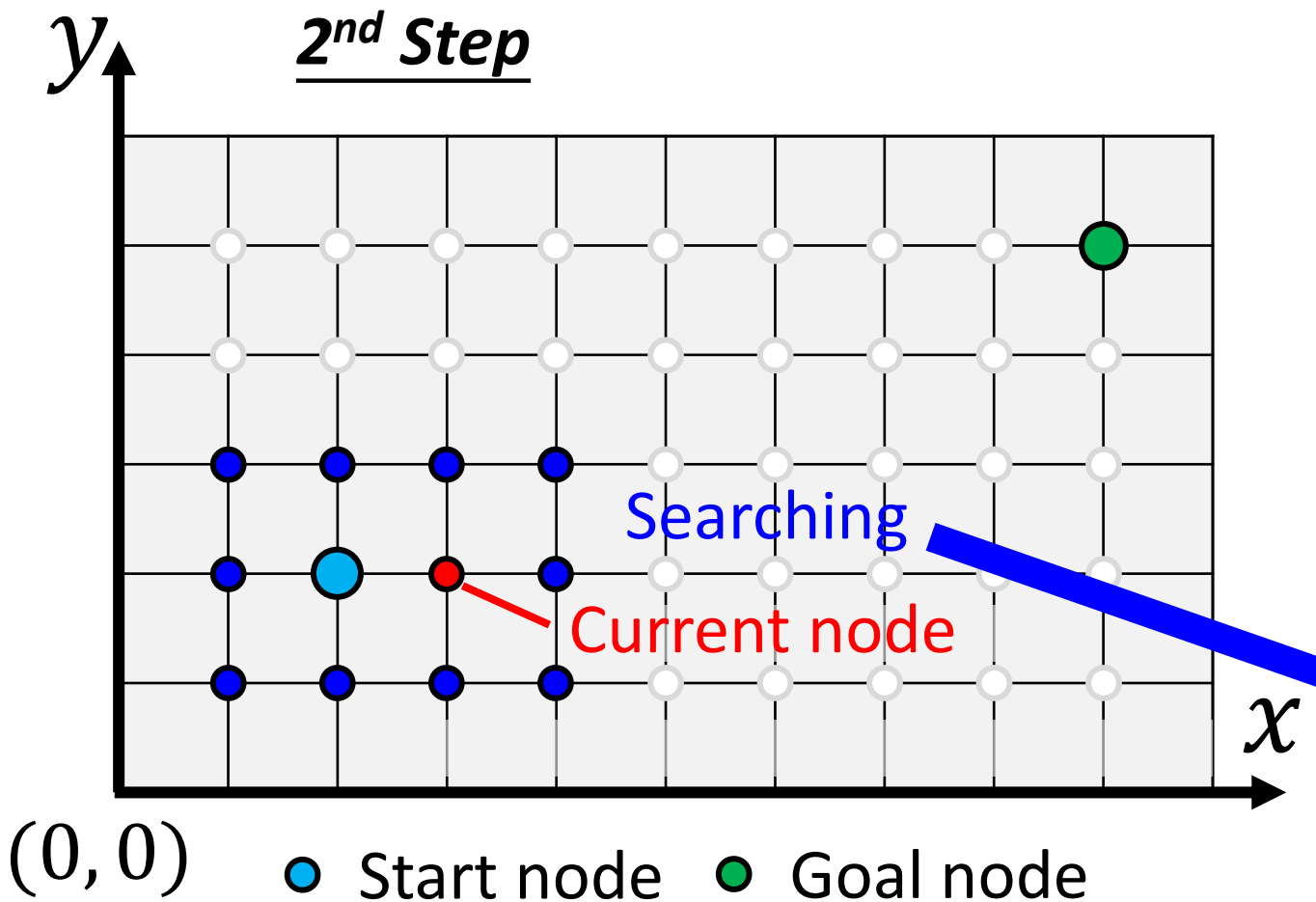
Open List *(searched nodes)*

Node	f	Source
(1,1)	8.9	(2,2)
(1,2)	8.5	(2,2)
(1,3)	8.2	(2,2)
(2,1)	8.1	(2,2)
(2,3)	7.3	(2,2)
(3,1)	7.2	(2,2)
~~(3,2)~~	~~6.7~~	~~(2,2)~~
(3,3)	6.3	(2,2)

Close List *(arrived nodes)*

Node	f	Source
Start	-	-
(3,2)	6.7	(2,2)

A-star (A*) Path Planning – Record Path



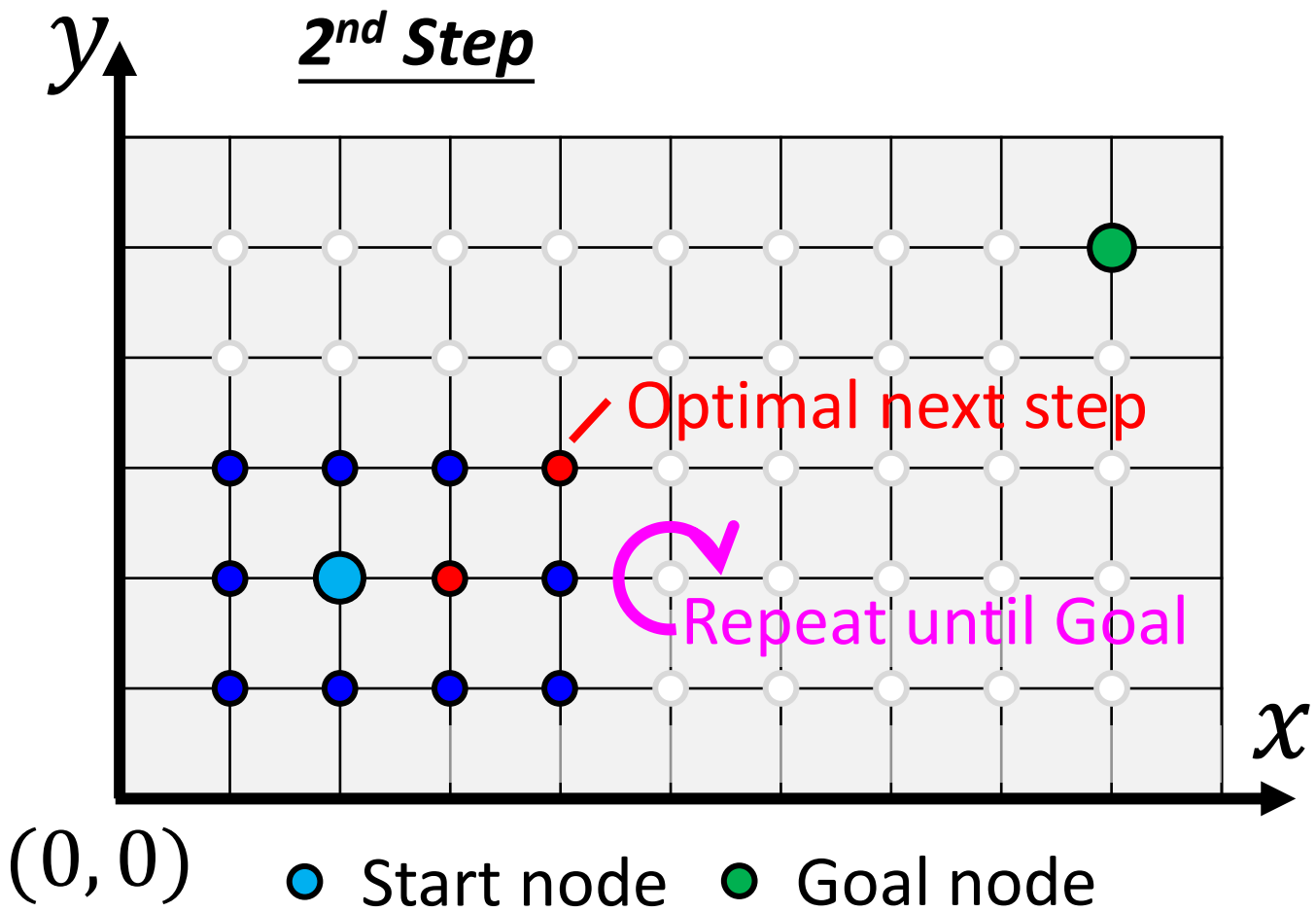
Open List *(searched nodes)*

Node	f	Source
(1,1)	8.9	(2,2)
⋮	⋮	⋮
(3,1)	7.2	(2,2)
(3,3)	6.3	(2,2)
⋮	⋮	⋮
(4,1)	7.8	(2,2)
(4,2)	7.8	(2,2)
(4,3)	8.8	(2,2)

Close List *(arrived nodes)*

Node	f	Source
Start	-	-
(3,2)	6.7	(2,2)

A-star (A*) Path Planning – Record Path



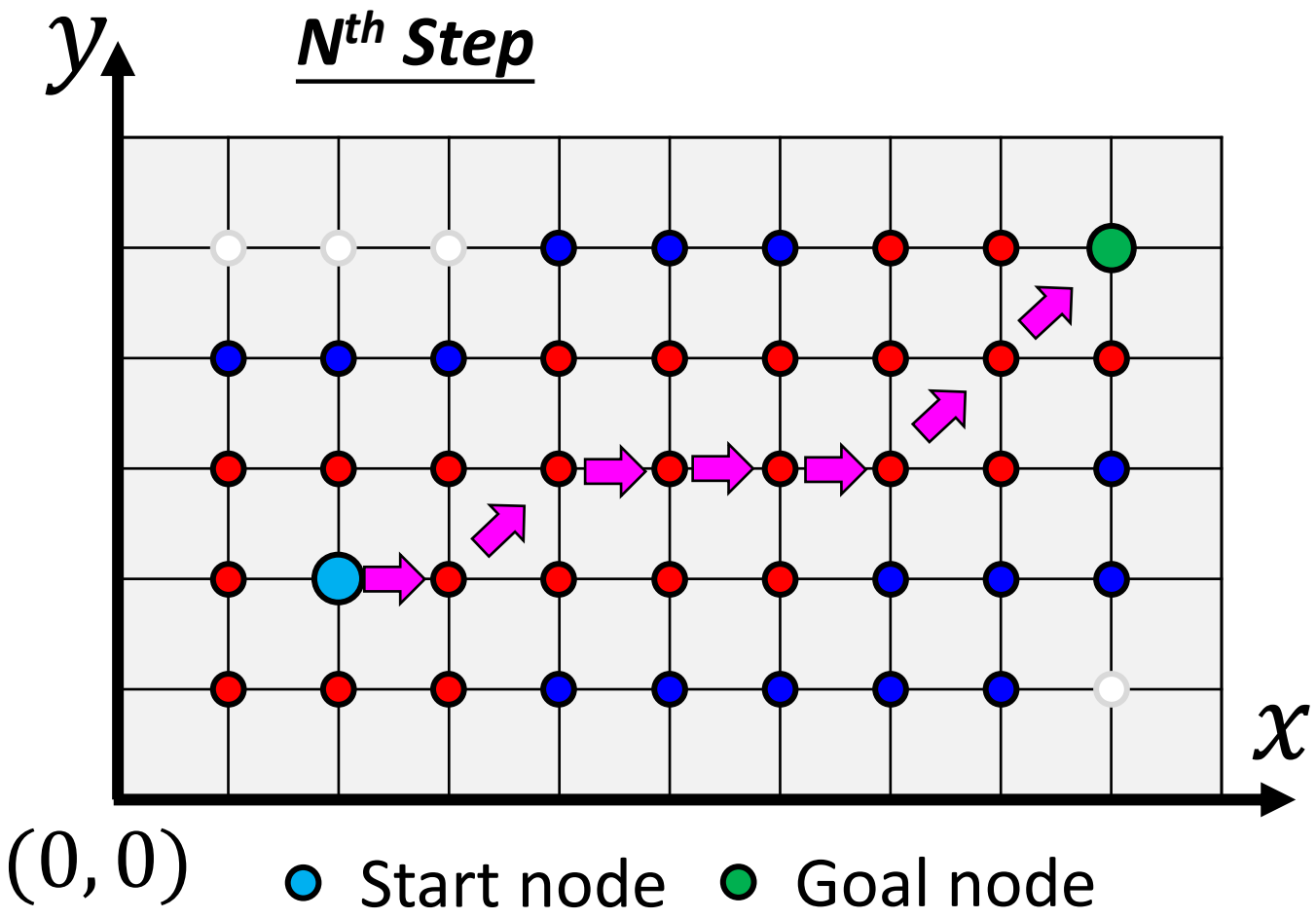
Open List *(searched nodes)*

Node	f	Source
(1,1)	8.9	(2,2)
⋮	⋮	⋮
(3,1)	7.2	(2,2)
(3,3)	6.3	(2,2)
⋮	⋮	⋮
(4,1)	8.8	(3,2)
(4,2)	7.8	(3,2)
~~(4,3)~~	~~7.8~~	~~(3,2)~~

Close List *(arrived nodes)*

Node	f	Source
Start	-	-
(3,2)	6.7	(2,2)
(4,3)	7.8	(3,2)

A-star (A*) Path Planning – Retrieve Best Path



Open List *(searched nodes)*

Node	f	Source
(1,1)	8.9	(2,2)
⋮	⋮	⋮
(3,1)	7.2	(2,2)
(3,3)	6.3	(2,2)
⋮	⋮	⋮
⋮	⋮	⋮
⋮	⋮	⋮
Goal	x	(x,y)

Close List *(arrived nodes)*

Node	f	Source
Start	-	-
(3,2)	6.7	(2,2)
(4,3)	7.8	(3,2)
⋮	⋮	⋮
⋮	⋮	⋮
Goal	x	(x,y)

Trace back arrived nodes

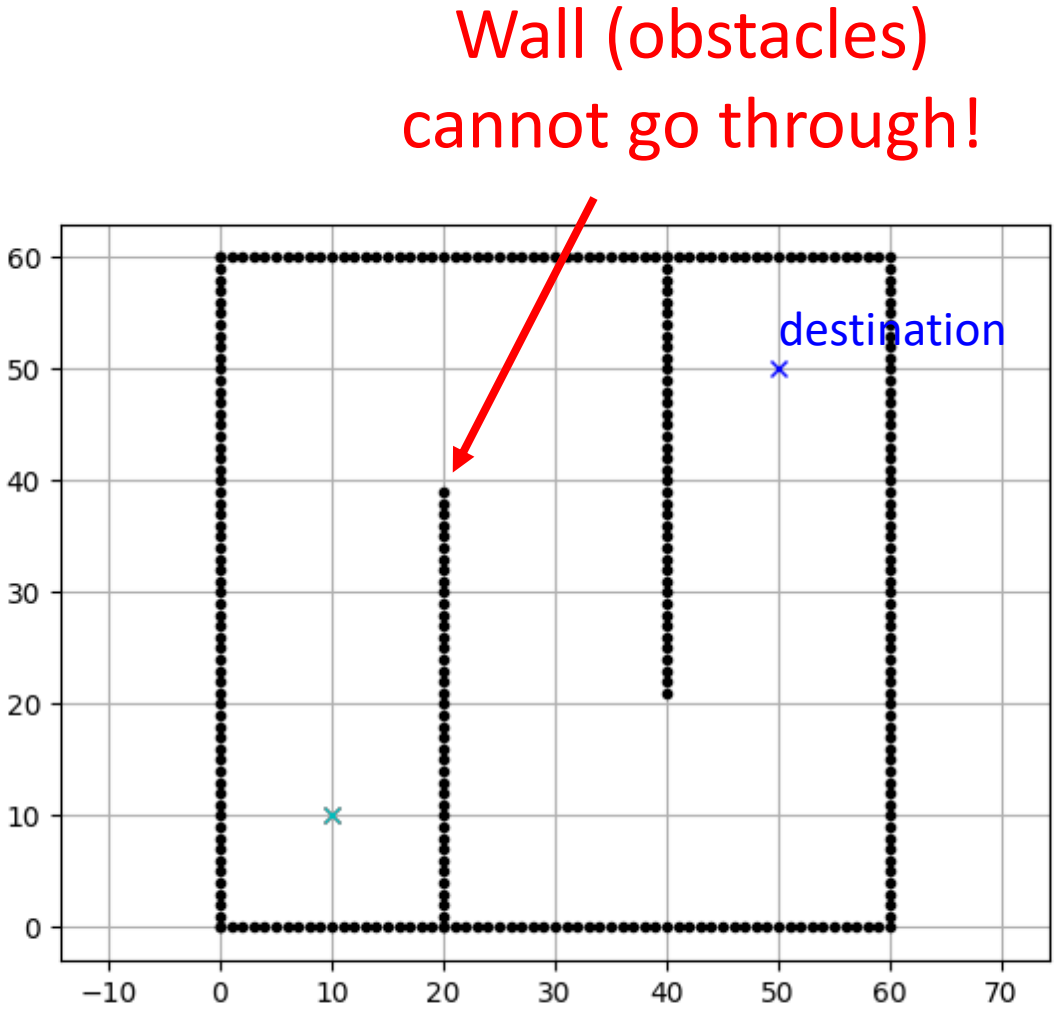
A-star Example

Each time A* enters a node, it calculates the cost: f(n) - n being the neighboring node

It travel to all of the neighboring nodes, and then enters the node with the lowest value of f(n)

These values we calculate using the following formula:

$$f(x,y) = g(x,y) + h(x,y)$$



https://towardsdatascience.com/a-star-a-search-algorithm-eb495fb156bb *Source: PythonRobotics*

Trip Cost of Flight

Trip Cost of Flight

The fundamental rationale of the cost index concept is to achieve minimum **trip cost** by means of a trade-off between operating costs per hour and incremental fuel burn.

Trip Cost: $$C = C_F \cdot \Delta F \cdot T + C_T \cdot T + C_c$$

* Related to travelling time

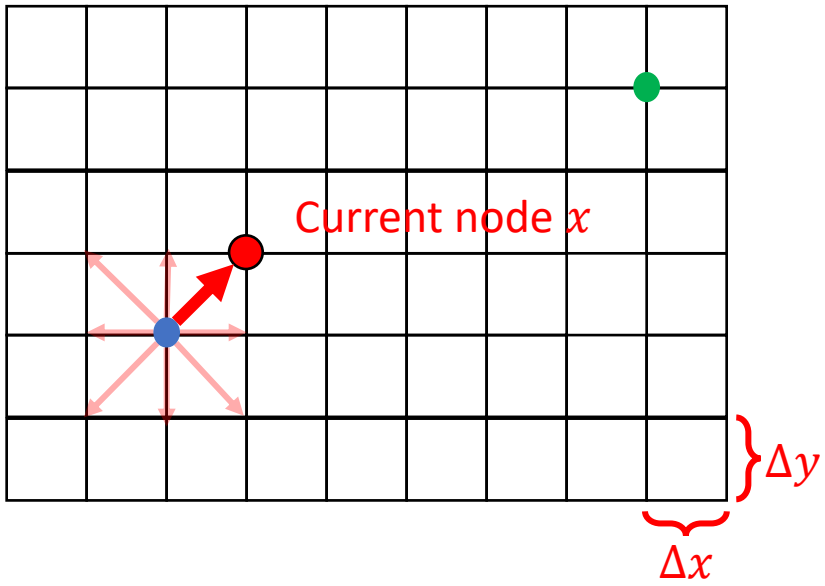
With

- C_F=cost of fuel per kg
- C_T=time related cost per minute of flight
- C_c=fixed cost independent of time
- ΔF=trip fuel (e.g., 3000kg/h)
- T=trip Time (e.g., 8 hours from Hong Kong to Paris)

Consider this relationship to imitate the path planning for flights?



Conventional Path Planning Task



Total cost: $f(x,y) = g(x,y) + h(x,y)$

Distance information (conventional)

Example for one step:

$$g(x,y) = \sqrt{\Delta x^2 + \Delta y^2}$$

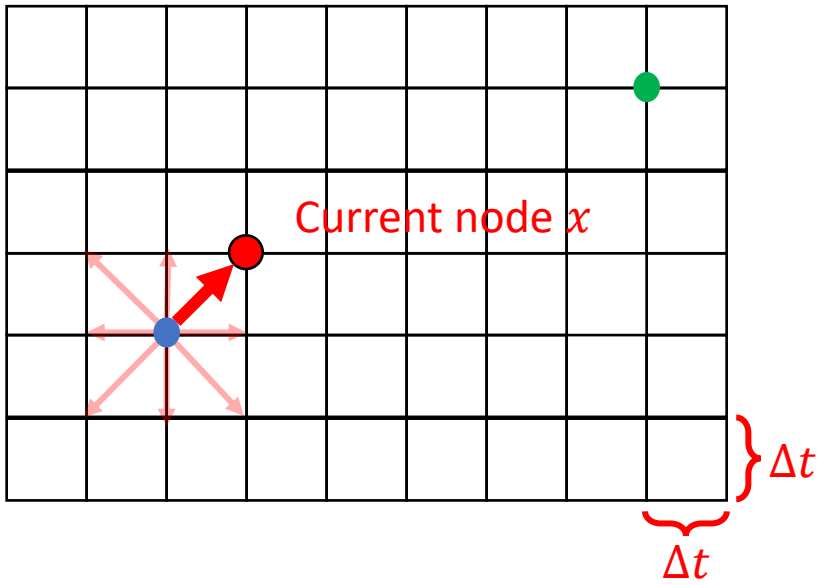
Objective: Find the path with the lowest traveling distance. (in the unit of meter)

Goal node Start node

Path Planning Task for Out Flight

Total cost: $f(n) = g(n) + h(n)$

Traveling time (our case for flight)



Time cost on each step:

$$s(n) = \begin{cases} \Delta t, & vertical/horizontal\ motion \\ \sqrt{2}\Delta t, & diagonal\ motion \end{cases}$$

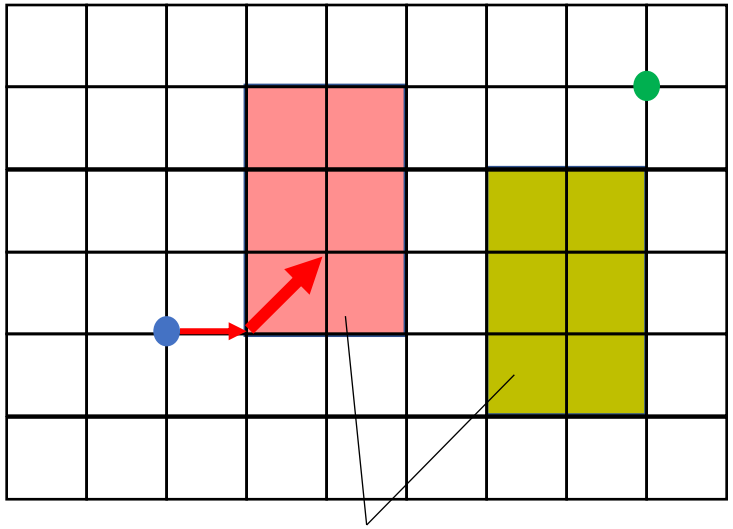
Example for first step:

$$g(n) = s(n) = \sqrt{2}\Delta t$$

Objective: Find the path with the lowest traveling time. (in the unit of min)

Goal node Start node n – index of nodes

Flight Planning Considering Cost Intensive Areas



Cost intensive area
(addition time)

Cost Intensive Areas: The cost for flying through such area is **increased** due to airflow, legal restrictions and other reasons.

Total cost for one node:

$$f(n) = g(n) + \alpha \cdot s(n) + h(n)$$

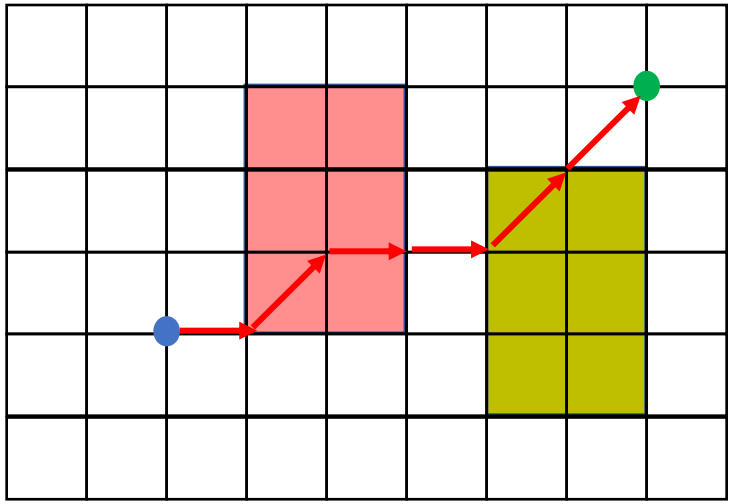
Previous cost from start

Additional time at current step

α – additional time factor (equal zero for normal area)

Goal node Start node

Flight Path Planning Trip Cost



Cost Intensive Areas: The cost for flying through such area is **increased** due to airflow, legal restrictions and other reasons.

Total cost for one node:

$$f(n) = g(n) + \alpha \cdot s(n) + h(n)$$

Path planning

$$T_{best} = \min[f(goal)]$$

Additional cost on each node

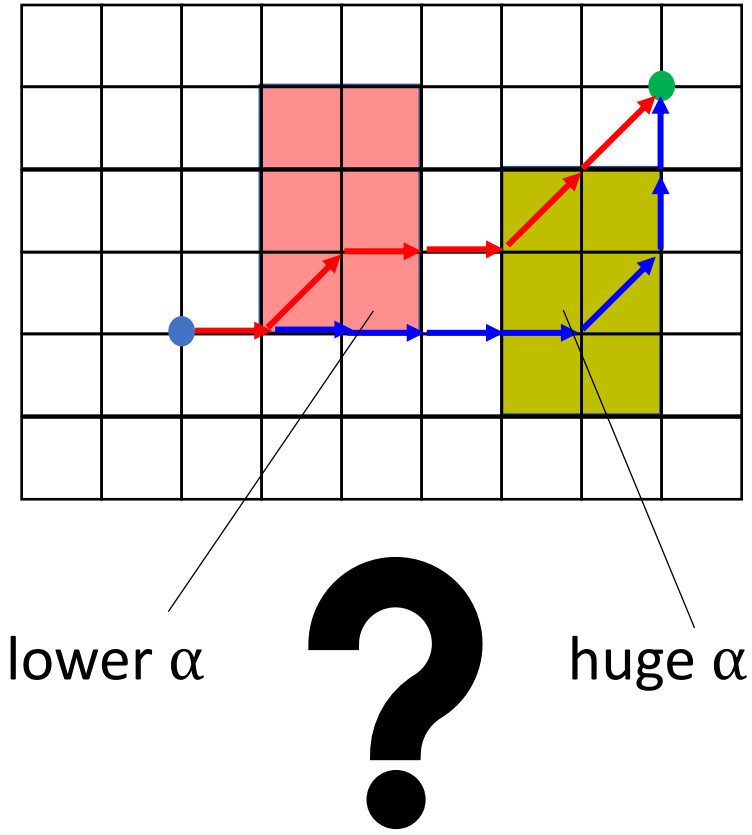
$$= \min\left[g(goal) + \alpha_1 s(n_1) + \alpha_2 s(n_2) + \cdots + \alpha_{goal} s(n_{goal})\right]$$

Trip cost for planned path:

$$C = C_F \cdot \Delta F \cdot T_{best} + C_T \cdot T_{best} + C_c$$

● Goal node ● Start node α – additional time factor

How it choose different routes?



Cost Intensive Areas: The cost for flying through such area is **increased** due to airflow, legal restrictions and other reasons.

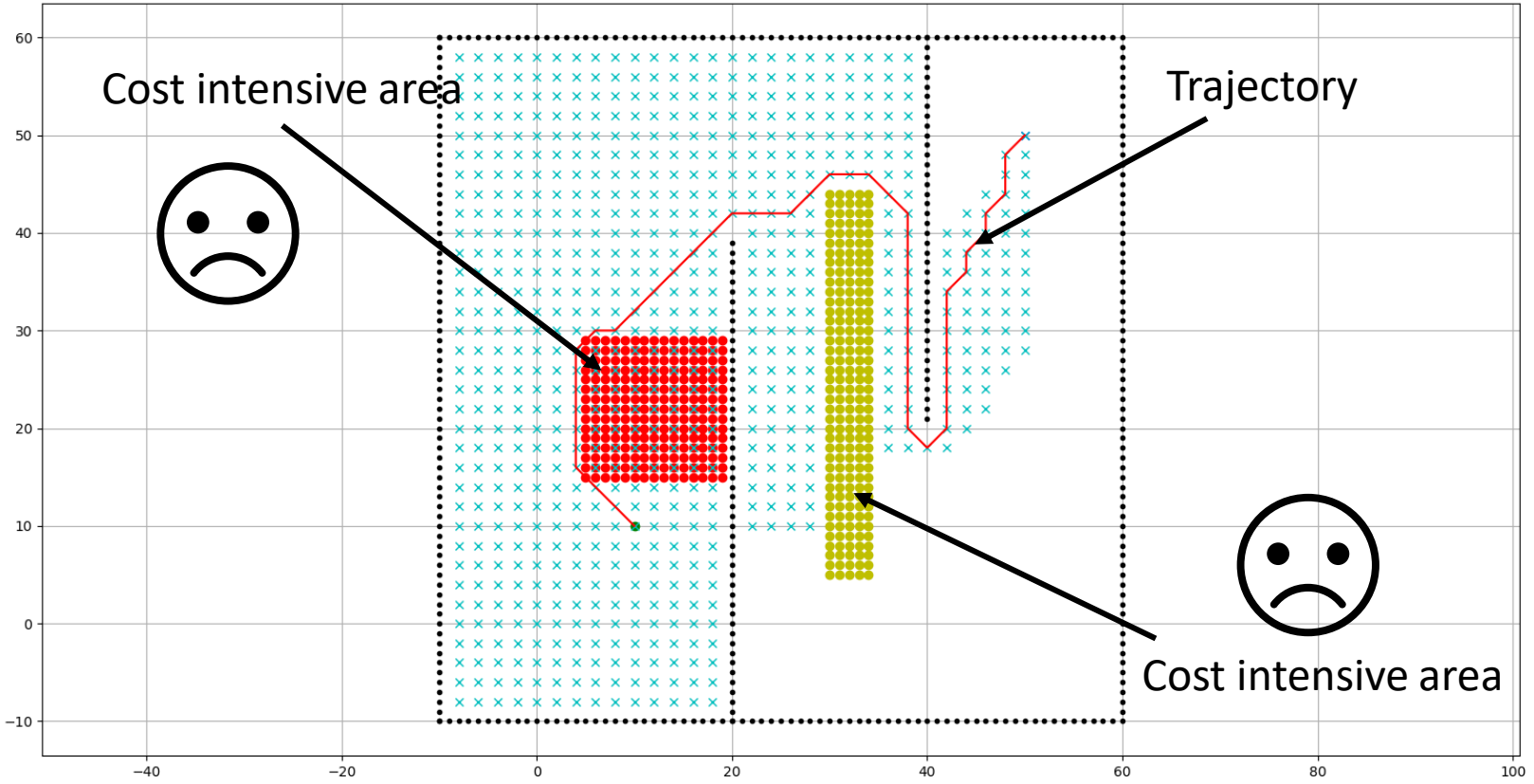
Time for the planned path:

$$T_{best} = \min\left[g(goal) + \alpha_1 s(n_1) + \alpha_2 s(n_2) + \cdots + \alpha_{goal} s(n_{goal})\right]$$

Depending on the extra time accumulated more specifically, the additional time factor

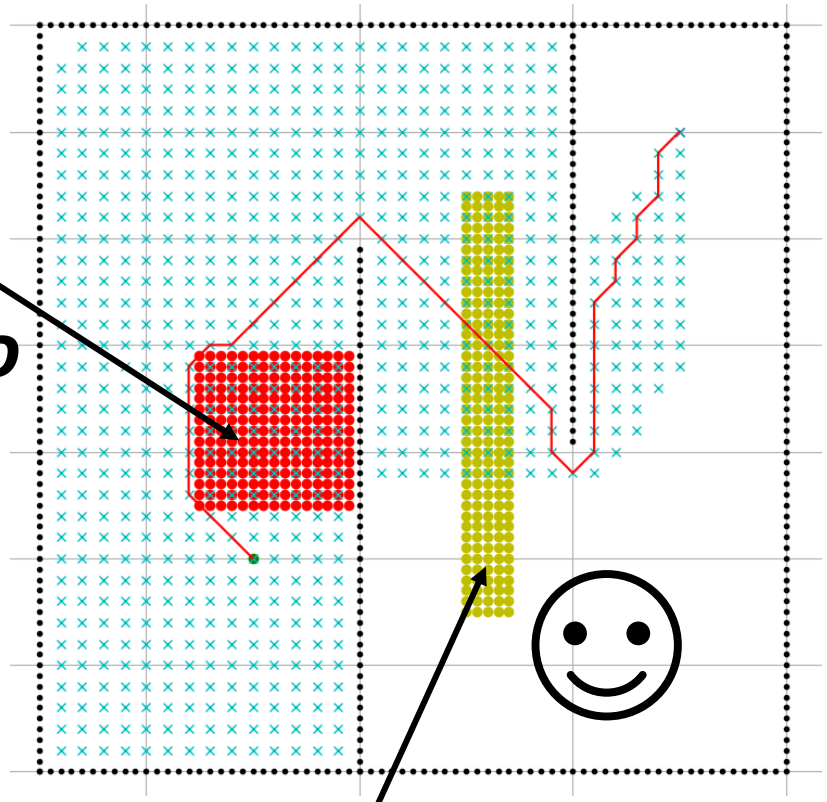
Goal node Start node α – additional time factor

Example of Flight Path Planning



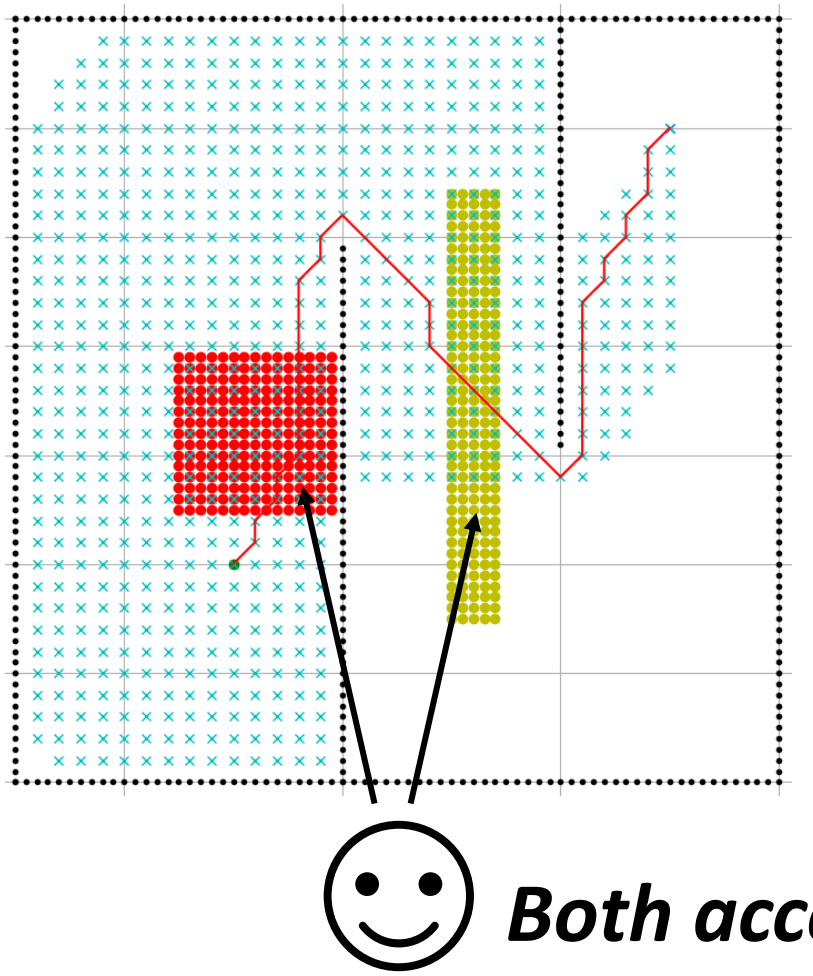
Cost is way too high for going through, better to avoid!

Example of Flight Path Planning



Cost too much

Not too much and acceptable



Both acceptable

Flight Path Planning Project



You will be creating and completing your own path planning program based on groups



You can find the project tasks / requirements in the GitHub homepage



Additional resources could be found inside the course GitHub repository

- Video tutorial
- Tutorial slides